Video-based Face Recognition Using Exemplar-Driven Bayesian Network Classifier

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***Abstract*—Many recent works in video-based face recognition involved the extraction of exemplars to summarize face appearances in video sequences. However, there has been a lack of attention towards modeling the causal relationship between classes and their associated exemplars. In this paper, we propose a novel Exemplar-Driven Bayesian Network (EDBN) classifier for face recognition in video. Our Bayesian framework addresses the drawbacks of typical exemplar-based approaches by incorporating temporal continuity between consecutive video frames while encoding the causal relationship between extracted exemplars and their parent classes within the framework. Under the EDBN framework, we describe a non-parametric approach of estimating probability densities using similarity scores that are computationally quick. Comprehensive experiments on two standard face video datasets demonstrated good recognition rates achieved by our method.**

I. Introduction

Machine recognition of faces has seen rapid developments in the past mainly in still image-based recognition, with a wide variety of state-of-art methods [1]. However, these methods tend to perform poorly in the presence of complex face variations under unconstrained environments. With the emergence of video, the abundance of images has presented a fast growing research area in video-based face recognition (VFR). A landmark psychological and neural study [2] have also reported that the human brain performs recognition of faces by means of both invariant structure of features and also idiosyncratic movements and gestures.

Many recent works [3], [4], [5], [6], [7] have been motivated to improve machine recognition of faces by exploiting temporal dynamics in video sequences. Some methods focused on directly modeling temporal dynamics by learning transitions between different appearance models. These methods tend to perform poorly under real-world conditions where facial variations are likely to be demanding. There also exist methods that represent a face manifold by extracting local models or exemplars, a set of representative images that summarizes a video sequence. Due to the reduction of features, these methods typically lack the ability to incorporate temporal continuity, especially in the classification step.

In this paper, we propose a new Exemplar-Driven Bayesian Network (EDBN) classifier for face recognition in video, which introduces a joint probability function that incorporates temporal continuity between consecutive video frames with consideration for the causal relationships between extracted exemplars and their respective parent classes. We first apply Locally Linear Embedding (LLE) [8] to learn a low-dimensional embedding of the training data before clustering with hierarchical agglomerative clustering. The exemplars are then extracted from the clusters for the recognition of new video sequences with the proposed classifier.

A. Related Work

Some recent approaches seek to exploit temporal dynamics by modeling transitions between appearance models. Zhou *et al.* [3] apply sequential importance sampling (SIS) to model a joint probability distribution of identity and head motion for simultaneous tracking and recognition. Lee *et al.* [4] approximate a nonlinear appearance manifold as a set of linear sub-manifolds, and transition probabilities were learned to model the connectivity between sub-manifolds. Due to their simplistic modeling of densities, these methods are highly sensitive to conditions where statistical correlations between test and training data are weak.

Methods that utilize Hidden Markov Models (HMM) to capture facial dynamics in video have also generated much interest in recent years [9], [10]. Despite showing promising results, the main disadvantage of these methods is that the learning of temporal dynamics during recognition can be computationally demanding, rendering these approaches almost infeasible for practical usage without further optimization and assumptions. Also, the problem of over-fitting during training is often a barrier to obtain a truly optimal parameter set needed for successful recognition.

Exemplar-based approaches have become increasingly popular in the literature. Krüeger and Zhou [11] proposed a method of selecting exemplars from face videos using radial basis function network. Hadid and Pietikäinen [5] proposed a view-based scheme which embeds the training face manifold using LLE algorithm, followed by applying k-means clustering in the embedding space. Cluster centers are extracted as exemplars and a probabilistic voting strategy is used to classify

new data. Fan *et al.* [6] used a similar configuration except that classification is performed using a Bayesian inference model to exploit temporal dynamics. Liu *et al.* [7] formulated a spatio-temporal embedding based on Bayesian keyframe learning and statistical classification. None of these methods consider the influence of different exemplars with respect to its class.

The major contribution of this paper is to introduce a joint probability function in a Bayesian network classifier which considers the relationship between exemplars and their classes. We also present a non-parametric approach to probability distributions using similarity scores that are computationally inexpensive.

II. Face Recognition in Video

A. Problem Setting

While conventional still image-based face recognition is a straightforward matching of a test image to a gallery of training images *i.e.* an *image-to-image* recognition task, it is an ill-posed problem for video sequences.

The configuration used by exemplar-based approaches to accomplish a complete *video-to-video* setting is by simplifying it to an *image-to-video* recognition task, where all appearances in each training video are summarized by a set of image exemplars [5], [12].

For the test video, it is intuitive to perform simple matching between each exemplar image and each frame from the test video. Ultimately, this classification assembly is aggregated using our Bayesian model (in Section 3) to achieve a representation of a full *video-to-video* recognition task.

For general notation, given a sequence of face images extracted from a video,

$$X_c = \left\{x_1^c, x_2^c, \ldots, x_{N_c}^c\right\} \ , \tag{1}$$

where N_c is the number of face images in the video. Assuming that each video contains the faces of the same person and c is the subject identity of a C-class problem, $c \in \{1, 2, \ldots, C\}$, its associated exemplar set

$$E_c = \{e_1^c, e_2^c, \ldots, e_M^c\} \ , \tag{2}$$

where $E_c \subseteq X_c$ and the number of exemplars extracted, $M \ll N_c$. Thus, the overall exemplar-class set can be succinctly summarized as

$$E = \{e_{c,m} | c = 1, \ldots, C; m = 1, \ldots, M\} \ , \tag{3}$$

in which there are a total of $C \times M$ unique exemplar-classes. In cases where more than one training video of a particular class is used, image frames from all similar-class videos are aggregated to extract M exemplars.

B. Modeling Face Variations

Considering the large amount of face variations in each training video sequence, we first apply the nonlinear dimensionality reduction method, Locally Linear Embedding (LLE) [8] to learn a low-dimensional embedding from its original data space. It is well known that LLE is capable of modeling the intrinsic structure of a nonlinear data manifold in a meaningful embedding to better capture various face variations such as pose, expression and illumination.

C. Extraction of Exemplar Set by Clustering

Next, the projected faces in LLE-space for each training video sequence are partitioned into clusters using hierarchical agglomerative clustering (HAC) [12], [13].

This clustering step aims to form groups of faces that contain strong correlation of appearances. In brief, the HAC algorithm is described as follows:

1) Initialize each data point (of all N_c points) as a singleton cluster.
2) Find the nearest pairs of clusters, according to a certain distance measure between clusters. Some commonly used measures are the single-link, complete-link, group-average-link and Ward's distance criterion [13]. The two nearest clusters are merged to form a new cluster.
3) Continue distance computation and merging (repeat Step 2), and terminate when all points belong to a single cluster. The required number of clusters, M is selected by partitioning at the appropriate level of the cluster tree.

The HAC method is chosen due to its clear advantages over the standard *k*-means clustering [14].

For each cluster, the face that is nearest to the cluster mean is selected as the *exemplar*, or a representative face image of the cluster (see Fig. 1). The final exemplar set consists of the extracted exemplars from all training video sequences. Fig. 2 shows some sample extracted exemplar sets from two datasets used in this work.

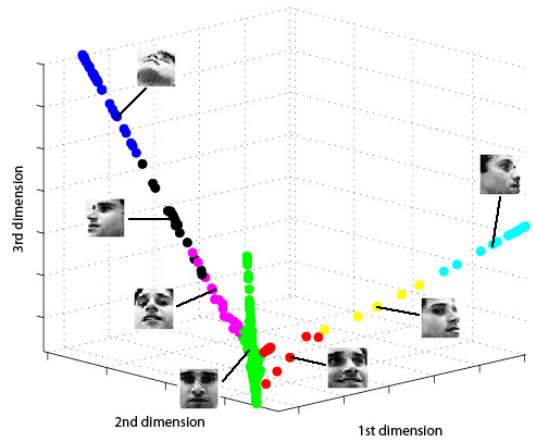
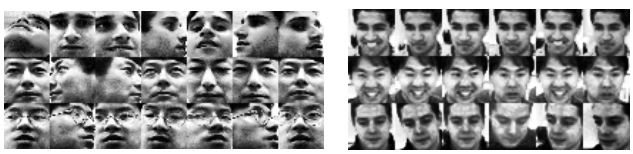


Fig. 1. Face exemplars extracted from clusters in LLE-space for a sample training sequence (each cluster shown in different colors).



(a) (b)

Fig. 2. Sample extracted exemplar sets (one exemplar set per row) of the (a) Honda/UCSD and (b) CMU MoBo datasets

III. Bayesian Network Framework for Recognition

In a Bayesian inference model, the subject identity of a test video X can be found by estimating the *maximum a posteriori* (MAP) probability decision rule,

$$c^* = arg \max_C P(c|x_{1,\ldots,N_c}), \tag{4}$$

where the subscript notation of x succinctly represents a sequence of N images.

In a typical Naive Bayes classifier, estimation based on MAP decision rule can be expressed as

$$\begin{aligned} P(c|x_{1,\ldots,N_c}) &= \frac{P(c)P(x_{1,\ldots,N_c}|c)}{P(x_{1,\ldots,N_c})} \\ &= \frac{P(c)P(x_{1,\ldots,N_c}|c)}{\sum_c P(x_{1,\ldots,N_c}|c)P(c)} \ , \end{aligned} \tag{5}$$

where $P(c)$ is the prior probability of each class, $P(x_{1,\ldots,N_c}|c)$ is the likelihood probability of x given class c and the denominator is a normalization factor that ensures that the sum of the likelihoods over all possible classes equals to 1. Assuming conditional independence between all observations *i.e.* $x_i \perp\!\!\!\perp x_j|c$ where $i \neq j$, Eq. (5) can be rewritten as,

$$P(c|x_{1,\ldots,N_c}) = \prod_{i=1}^{N_c} \frac{P(c)P(x_i|c)}{P(x_i)} \ . \tag{6}$$

We propose an Exemplar-Driven Bayesian Network (EDBN) framework by introducing a joint probability function,

$$P(c, E, X) = P(X|c, E)P(E|c)P(c) \ , \tag{7}$$

where the exemplar-class set E is a new latent variable. The graphical model of the EDBN classifier is shown in Fig. 3.

Thus, the MAP classifier is redefined by maximizing the *joint posterior probability* of the class c and exemplar-class E given observation X:

$$\begin{aligned} \max_C P(c, E|X) &= \max_C \frac{P(c, E, X)}{P(X)} \\ &= \max_C \sum_{j=1}^{M} \frac{P(X|c, e_{c,j})P(e_{c,j}|c)P(c)}{P(X)} \\ &= \max_C \sum_{j=1}^{M} \prod_{i=1}^{N_c} \frac{P(x_i|c, e_{c,j})P(e_{c,j}|c)P(c)}{P(x_i)} \ . \end{aligned} \tag{8}$$

Intuitively, the conditional probability $P(e_{c,j}|c)$ acts as an *exemplar prominence* weight for the class likelihood $P(x_i|c, e_{c,j})$. The marginal probability $P(x_i)$ does not depend on both c and E, thus functioning only as a normalization constant. Since the class prior probability $P(c)$ is assumed to be non-informative at the start of observation sequence X, uniform priors are a good estimation.

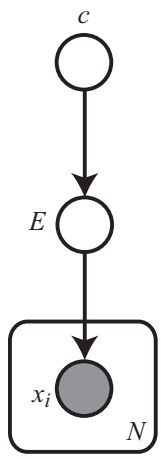


Fig. 3. Graphical model of the proposed EBN classifier

A. Computation of Likelihood $P(x_i|c, e_{c,j})$

In conventional Bayesian classifiers, a multivariate Gaussian density function is used to estimate data distribution. However with the limited sample size in our problem setting, accurate estimation of distribution can be challenging and easily result in over-fitting or under-fitting of data. An alternative method of performing non-parametric density estimation is by using distance measures (or a kernel density estimator with uniform kernel), which are computationally inexpensive.

We define a Frame Similarity Score (FSS) as the reciprocal of the ℓ^2-norm distance between the observed face image x_i and the j-th exemplar-class $e_{c,j}$,

$$S_i^{FSS}(x_i, e_{c,j}) = \frac{1}{d_{\ell^2}(x_i, e_{c,j})} \ . \tag{9}$$

The likelihood of the test face image x_i given the class c and exemplar-class e is determined by the ratio of FSS for exemplar-class $e_{c,j}$ to the total sum of FSS across all $C \times M$ exemplar-classes,

$$P(x_i|c, e_{c,j}) = \frac{S_i^{FSS}(x_i, e_{c,j})}{\sum_{k=1}^{C} \sum_{m=1}^{M} S_i^{FSS}(x_i, e_{k,m})} \ . \tag{10}$$

B. Computation of Exemplar Prominence $P(e_{c,j}|c)$

Causal relationship between exemplars and their respective classes can be represented by the exemplar prominence probability $P(e_{c,j}|c)$. Similar to the computation of likelihood probability, we avoid estimating density functions by representing the influence of an exemplar in its own class subspace by a normalized Hausdorff distance metric [15],

$$d_h(e_{c,j}, E_c) = \frac{1}{\Lambda} \min_{e' \in E_c} \|e_{c,j} - e'\| \ , \tag{11}$$

where E_c is the exemplar set of class c and Λ is a sum normalization factor to ensure distances are relative within-class.

By defining the Exemplar Prominence Score (EPS) as the reciprocal of the distance metric,

$$S_{c,j}^{EPS}(E_c, e_{c,j}) = \frac{1}{d_h(e_{c,j}, E_c)} \ , \tag{12}$$

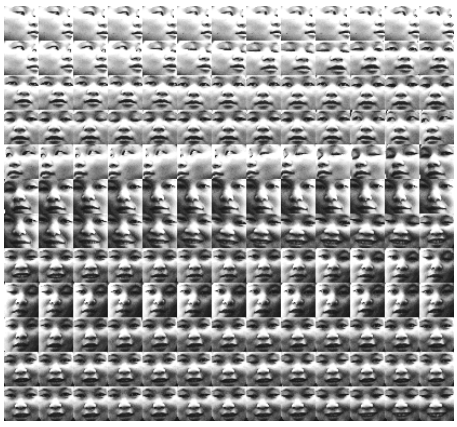
the exemplar prominence probability can be formulated as

$$P(e_{c,j}|c) = \frac{S_{c,j}^{EPS}(E_c, e_{c,j})}{\sum_{m=1}^{M} S_{c,j}^{EPS}(E_c, e_{c,m})} \ , \tag{13}$$

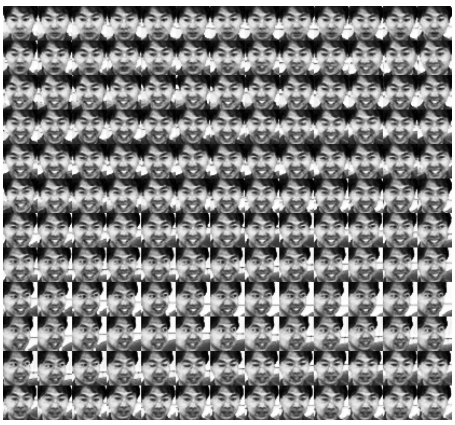
which can be pre-computed offline since it does not depend on input observation X.

IV. Experiments and Discussion

To ensure an extensive evaluation of different classification methods, comprehensive experiments were conducted on two standard video face datasets: the Honda/UCSD [4] and CMU MoBo [16] (see Fig. 4).



(a) Honda/UCSD



(b) CMU MoBo

Fig. 4. Sequentially-ordered image frames of a sample video sequence taken from the evaluated datasets

The first dataset, Honda/UCSD, which was collected specifically for video-based face recognition, consists of 59 video sequences of 20 different people (each person has at least 2 videos). Each video contains about 300-600 frames, comprising of large pose and expression variations with significantly complex out-of-plane (3-D) head rotations. The second dataset, CMU MoBo is a widely used benchmark dataset for video-based face recognition. It consists of 96 sequences of 24 different subjects (each person has 4 videos). Each video contains about 300 frames.

For each video sequence, faces are extracted using the Viola-Jones cascaded face detector [17] and resampled to 32×32 pixels. Histogram equalization was applied to normalize lighting effects. Sample exemplar images from both datasets were shown earlier in Fig. 2.

In our experiments, we use one video sequence for training and the remaining video sequences for testing. To evaluate all subjects in each dataset extensively, the test sequence set is constructed by randomly sampling 20 subsequences consisting of 100 frames each from every test video. We fix the number of extracted exemplars, $M = 7$ for Honda/UCSD and $M = 6$ for CMU MoBo. The common rule of thumb for determining the value of M is by identifying the "elbow" or trade-off point of the residual error curve from the earlier clustering step [13]. determining the value of M is by identifying the "elbow" or trade-off point of the residual error curve from the earlier clustering step [13].

The choice of dimensionality reduction method is the state-of-the-art nonlinear method, Neighborhood Discriminative Manifold Projection (NDMP)[18], which is able to extract meaningful discriminative features from a highly nonlinear data manifold of video sequences. The face images from both exemplar sets and test sets are projected to the NDMP-space before the classification task.

By applying the proposed EDBN classifier in the exemplar-based setting, a *video-to-video* recognition task is accomplished by maximizing Eq. (8) across all frames of a test sequence to decide on the subject identity. In Fig. 5, the posterior plot of a sample test sequence from the Honda/UCSD dataset demonstrated that the proposed Bayesian recognition framework is capable of arriving at the correct identity, even when the initial frames were incorrectly classified.

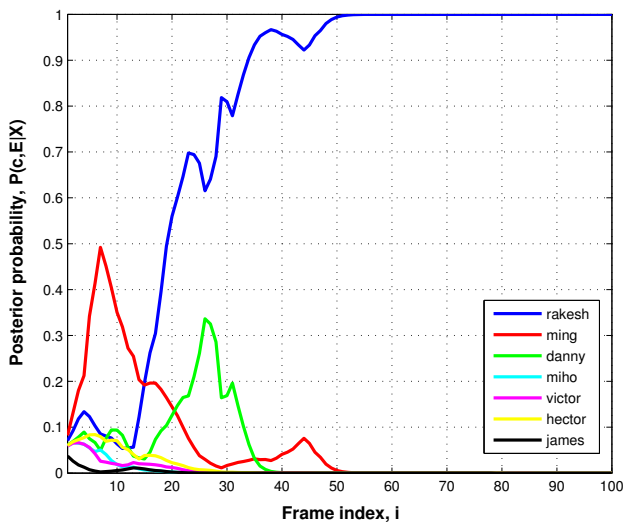


Fig. 5. Plot of posterior probability $P(c|E, X)$ versus frame index plot of the *rakesh* test video from Honda/UCSD dataset. Posteriors of the seven most probable subjects are shown in different colors. The subject (blue line) is correctly identified.

A. Comparative Results

To evaluate the effectiveness of classifying video sequences, we compare the performance of the following classification schemes on both Honda/UCSD and CMU MoBo datasets:

1) **Majority voting** (with nearest neighbor matching), where a vote is taken in each frame and the class with the majority vote is classified as the subject.
2) **Probabilistic voting**, where the likelihood probabilities of each frame (based on Eq. (10)) are combined cumulatively by simple sum rule. The class with the largest sum of likelihoods is classified as the subject.
3) **Naive Bayes classifier** (based on Eq. (5))
4) **Exemplar-Driven Bayesian Network classifier** (based on Eq. (8))

Table I shows the recognition rates on the Honda/UCSD and CMU MoBo datasets using the methods listed above. The results show that Bayesian classifiers generally perform better in the video-based setting where rapidly changing face pose and expression can easily cause recognition failure. Unlike traditional voting strategies, temporal dependencies between video frames are well-established in our framework. Compared to the Naive Bayes classifier, additional causality between exemplars and their respective parent classes in the EDBN classifier further enhances recognition accuracy as exemplars that are more prominent are given more influence in the inference model, and vice versa.

Table II shows the consistency of the EDBN classifier, where it yields higher recognition rates even when different features (PCA [19], LDA [20], NPE [21]) are used.

TABLE I
RECOGNITION RATES (%) OF CLASSIFICATION METHODS ON EVALUATED DATASETS

Classification method	Datasets	
	Honda/UCSD	CMU MoBo
Majority vote (with NN)	63.38	94.65
Probabilistic vote	70.78	95.08
Naive Bayes classifier	84.61	96.30
EDBN classifier	86.90	97.70

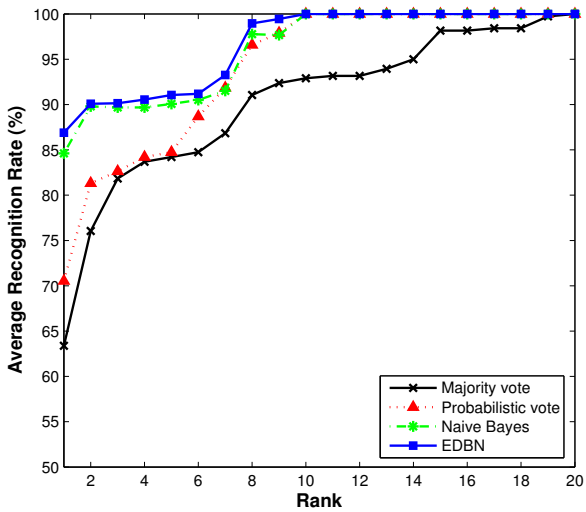
TABLE II
RECOGNITION RATES (%) WITH VARIOUS SETTINGS ON THE CMU MOBO DATASET

Classification method	Feature			
	PCA	LDA	NPE	NDMP
Majority vote (with NN)	83.28	90.95	93.53	94.65
Probabilistic vote	82.16	92.93	92.33	95.08
Naive Bayes classifier	85.63	93.66	93.27	96.30
EDBN classifier	86.44	94.97	94.11	97.70

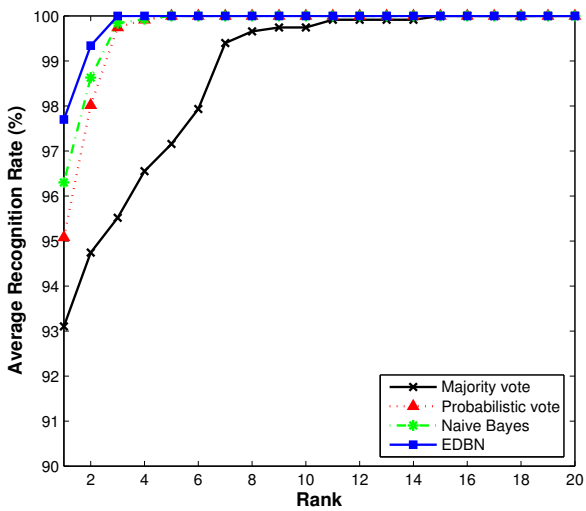
B. Rank-based Identification

We further evaluate the reliability and robustness of the proposed classifier in a rank-based identification setting. To accommodate this setting, we alter the MAP decision rule in (4) to take the top-n matches (instead of maximum) based on the computed posterior probabilities. The subject in the test sequence is identified if it matches any class from the top-n matches.

Comparison between the performance of different classification methods are presented using a Cumulative Match Characteristic (CMC) curve, as shown in Figs. 6a and 6b. The CMC plots show that the EBDN classifier consistently yield better recognition rates throughout rank-n top matches for the Honda/UCSD. It also achieved a perfect recognition of all evaluated video sequences (100%) for the CMU MoBo dataset within the top 3 matches. Generally, the plots also demonstrated the effectiveness of Bayesian probabilistic methods, which is far more robust than traditional voting methods which merely serve as an aggregation of *image-to-image* classification.



(a) CMC curves for Honda/UCSD



(b) CMC curves for CMU Mobo

Fig. 6. Cumulative match characteristic curves for both evaluated datasets

C. Algorithm Complexity

In terms of computational complexity, the EDBN classifier has a time complexity of $\mathcal{O}(nm)$ compared to $\mathcal{O}(n)$ for a typical Naive Bayes classifier, where n is the number of frames in a video sequence and m is the number of exemplars per class. In our problem setting, this difference is insignificant as $m \ll n$ under typical conditions. Also, the exemplar-class representation (established in Section II-A) results in $\mathcal{O}(nm)$ for Naive Bayes since the number of "classes" is effectively nm. The computation of exemplar prominence values is the primary drawback of the $\mathcal{O}(cm)$ space complexity of EDBN (where c is the number of classes) but that can be computed offline during the training phase.

V. Conclusion

In this paper, we present a novel Exemplar-Driven Bayesian Network (EDBN) classifier for face recognition in video, which introduces causal relationships between extracted exemplars and their respective parent classes, while incorporating temporal continuity between consecutive video frames. In our extensive experiments on standard datasets, the EDBN achieved better recognition rates compared to conventional methods. Also, the robustness of the EDBN classifier is demonstrated by its consistent performance using different features. In future, further tests can be conducted to test the capability of EDBN in dealing with real-world scenarios such as multiple identities in a sequence, and degraded low-quality videos. Also, the performance of the current Bayesian model can possibly be improved by extending our scheme to accommodate image sets.

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